Constructive Language Locking — Separating Cognitive Freedom from Linguistic Protocol

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Why grammar cannot be discovered by physics, and what to do instead.

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Constructive Language Locking (CLL): Separating Cognitive Freedom from Linguistic Protocol

> *"Grammar is not a law of nature awaiting discovery. It is a protocol two people agreed to. Confusing the two is why physics-based language generation has failed."*

Abstract

Wave-based cognitive substrates have repeatedly failed to produce grammatical natural language, and the field's response has been to abandon the mechanical problem in favour of a statistical one: $P(w_n \mid w_1 \ldots w_{n-1})$. We argue this was a false dilemma created by a category error. Cognition and language are different kinds of process — one *discovers*, the other *transmits* — and attempting to derive grammar from field dynamics conflates a communication protocol with a physical phenomenon. We propose **Constructive Language Locking (CLL)**: the cognitive substrate remains entirely unconstrained and constructive, while surface realisation is delegated to *isolated language clusters* holding fixed, deterministic grammatical rules. The core loses no flexibility because no grammar rule ever enters it; the output gains exact, non-hallucinating surface form because nothing is sampled. We show the architecture emits one cognitive state into English, Bengali and Banglish from a single triple, at $O(1)$ lookup cost, with zero learned parameters.

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1. The Category Error

1.1 Two Processes, Wrongly Merged

	Cognition	Language
Purpose	discover what is true	transmit what is held
Requires	novelty, experiment, resonance, oracle validation	a fixed protocol both parties know
Bound by fixed rules?	never — that would cripple it	entirely — that is what it *is*
Failure mode	stagnation	unintelligibility
Correct source	physics	convention

English SVO is not a fact about the universe. It is a convention; Bengali SOV encodes identical cognitive content under a different one. No experiment discovers that English places verbs medially, because there is nothing there to discover — only something to *agree*.

1.2 Why This Killed Physics-Based Generation Earlier work in this lineage (`synthetic-intelligence.md` §5.4b, `mrs.md`) attempted to derive word ordering from field dynamics: syntax slots with fixed natural frequencies, phase-locked readout, gauge offsets enforcing subject-before-object. The concept-level ordering partially succeeded. The *byte*-level emission babbled.

The diagnosis was repeatedly treated as a tuning problem. It was not. Asking a resonance field to discover that English uses articles is asking it to derive a convention from a phenomenon. There is no gradient to follow because there is no fact of the matter.

The field ran to language models for the same reason. The statistical shortcut is not a theory of language production; it is what remains once the mechanical problem is abandoned. A next-token distribution requires no account of *why* one word follows another, only a count of how often it did. It works spectacularly, and it forecloses the question.

CLL claims the mechanical route was never seriously attempted because the problem was mis-stated, not because it is closed.

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2. The Locking Principle

Definition (Constructive Language Locking). A cognitive architecture satisfies CLL when:

1. **Core freedom.** No grammatical rule appears anywhere in the cognitive substrate. Its dynamics, novelty mechanisms, and structural operations are identical whether the output language is English, Bengali, or none.
2. **Protocol determinism.** Surface realisation is a total function from (cognitive state, language cluster) to string. The same input always yields byte-identical output. Nothing is sampled.
3. **Isolation.** Language clusters may be added, removed, or replaced without modifying the core. Their number does not affect cognitive capability.

The name records the asymmetry: cognition stays **constructive** (free to build), while language is **locked** (fixed to a protocol).

Corollary (No cognitive degradation). Because (1) forbids grammar in the core, adding a language cluster cannot reduce cognitive flexibility. This is a structural guarantee, not an empirical claim.

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3. Architecture

```
[ Core SI Physics Substrate ]
  resonance, phase-locking, fission, L6-A bridging,
  thermodynamic discovery, oracle validation
                |
                |  emits raw concept triples
                |  (subject, relation, object)
                v
[ Isolated Language Clusters ]
  +- English   : SVO grid, articles, 3sg agreement
  +- Bengali   : SOV grid, no articles, danda terminator
  +- Banglish  : SOV grid, Latin script, strict transliteration
                |
                v
        human-readable stream
```

The interface between the layers is deliberately narrow: a triple of graph labels plus a relation identifier. Nothing linguistic crosses upward, and nothing cognitive crosses downward.

3.1 Cluster Contents A language cluster is **data, not code**:

Component	Role	Size
Word order	SVO / SOV / VSO	~2 bits
Relation lexicalisation	`part_of` → "is part of"	1 row per relation
Determiner policy	per-construction, not per-language	1 flag per relation
Morphology rules	agreement, plural, conjugation	1 rule per pattern

Component	Role	Size
Transliteration map	তুমি ↔ tumi	1 row per word pair
Terminator	`.` / `।`	1 char

Adding a language means adding a table. The substrate is never touched.

3.2 Realisation Given triple (s, r, o) and cluster L:

$$\text{realise}_L(s, r, o) = \Pi_L(\lambda_L(s),\ \rho_L(r),\ \delta_L(r, o))$$

where λ_L is concept surface form (transliteration if any), ρ_L is relation lexicalisation, δ_L applies determiner policy conditioned on the *construction* r, and Π_L permutes by word order. Every component is a table lookup or a fixed string operation. The function is total and deterministic.

3.3 Measured Output One cognitive state, three protocols (`src/lang.hpp`):

```
(biology, is_a, science)      en: Biology is a science.
                              bn: Biology science একটা।
                              bl: Biology science ekta.

(metformin, inhibits, mtor)   en: Metformin inhibits mtor.
                              bn: Metformin mtor বাধা দয়ে।
                              bl: Metformin mtor badha dey.

(drive, precedes, tick)       en: Drive must run before tick.
                              bn: Drive tick আগে চলে।
                              bl: Drive tick age chole.
```

---

4. A Defect That Testing Disproved Immediately

The first implementation treated determiner insertion as a **global language property** — "English uses articles" — and produced:

> *"Biology is a the science."*

Whether an object takes a determiner is not a property of the language. It belongs to the **construction**: `is_a` already supplies its own article; proper nouns take none. The policy is now a per-relation flag.

This is recorded because it establishes the shape of all remaining work in this layer. Real grammar exceeds word order, and each additional rule is discovered by *testing against a speaker's judgement*, not by gradient descent. Every one is deterministic data.

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5. Transliteration as Strict Relation

Code-switching is where language models fail most visibly and CLL succeeds most cleanly. The mapping

$$\texttt{tumi} \xleftrightarrow{\text{REL_PHONETIC_EQUIV}} \text{তুমি}$$

is a **strict static relation**, not a probable correspondence. There is no distribution over spellings, so there is no drift. A language model generating Banglish must sample each character sequence and will produce *tumi*, *tomi*, *tumii* across a single document. CLL cannot, because nothing is sampled.

This generalises to dialect: Sylheti, Chittagonian and standard Bengali differ in phonotactic mapping while sharing cognitive content. Swapping the transliteration table swaps the dialect. The thought is unchanged.

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6. Cost Accounting

CLL introduces **no hidden cost** and must be honest about the cost it does introduce.

The Translation Capacity Lemma (`synthetic-intelligence.md` §3.1) proves that discriminating N semantic outcomes requires $\log_2 N$ bits somewhere. Layer 3 already pays this by storing label strings. CLL adds:

Addition	Cost	Category
Word order	2 bits per language	protocol
Relation lexicalisation	$O(\|\mathcal{R}\|)$ rows per language	protocol
Transliteration	$O(\|\text{lexicon}\|)$ rows per script	protocol

All are Layer-3 protocol cost, explicitly named. **None is intelligence.** A word-order constant does not help the system reason; it helps the system be understood.

Critically, the cost is $O(|\mathcal{R}|)$ in the number of *relations*, not $O(N^2)$ in concept pairs — because grammar generalises over relation types, which is exactly what makes it a protocol rather than a lookup table of sentences.

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7. Comparison

Property	Language model	CLL
Grammar source	inferred from token co-occurrence	stated convention
Sequencing	argmax over learned distribution	table permutation
Surface form stability	sampled — drifts	exact — cannot drift
New language	retrain or scale	add a table
Dialect switch	prompt and hope	swap transliteration map
Cost per sentence	GPU forward pass	hash lookup + concat
Failure mode	fluent hallucination	explicit gap
Cognitive coupling	grammar entangled in weights	zero — core untouched

The last row is the load-bearing one. In a language model, grammatical competence and world knowledge occupy the same parameters and cannot be separated. Under CLL they are different files.

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8. What Remains Open

CLL solves *protocol*. It does not solve *morphological productivity*.

Solved. Word order, determiner policy, relation lexicalisation, script and transliteration, terminator conventions, dialect swap.

Deterministic but unwritten. Subject-verb agreement (*studies*, not *study*); pluralisation; Bengali verb conjugation by person and honorific (তুমি / আপনি / তুই); case marking under free word order; multi-triple discourse with pronoun resolution. All are tables. None needs ML. They are labour, not research.

Genuinely open. Producing the bytes of a word **not in the lexicon** — inflecting an unseen stem, coining a compound. This is the residual of the emission problem (`synthetic-intelligence.md` §8.5), and CLL narrows it rather than solving it: for any word already in Layer 3, no byte synthesis is required at all. The open problem is morphological *productivity*, not speech.

Not attempted. Prosody, register, politeness gradation beyond pronoun choice, and anything requiring modelling the listener.

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9. Failure Modes

1. **Protocol drift from reality.** A hand-written grammar table can be simply wrong about the language. There is no gradient to correct it — only a native speaker. This is a genuine disadvantage relative to corpus-fitted models, and the correct mitigation is a held-out judgement set per language, not more rules.
2. **Coverage cliff.** A relation with no lexicalisation entry falls back to rendering its identifier. The output stays intelligible but degrades visibly. This is preferable to inventing a form, but it is a cliff rather than a slope.
3. **The template ceiling.** Single-triple realisation cannot produce subordination, relative clauses, or discourse cohesion. Extending to multi-triple output requires a discourse layer that does not yet exist, and it is not obvious that it remains purely tabular.
4. **False economy on free-word-order languages.** Bengali permits reordering because case marking carries role. A fixed SOV template is an approximation that will read stiffly, and the fix is case morphology — item (2) of §8, not a new mechanism.

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10. Summary

Physics-based language generation failed because it asked a resonance field to discover a convention. Constructive Language Locking separates the two: cognition remains fully constructive and unconstrained, while surface realisation is a deterministic protocol held in isolated, swappable language clusters.

The core loses nothing — a structural guarantee, since no grammar rule enters it. The output gains exactness that sampling architectures cannot have. And the mechanism costs a table per language against a GPU forward pass per sentence.

The remaining hard problem is smaller and better specified than before: not *how does a field speak*, but *how does a field inflect a word it has never seen*.